XMAKE • GLASS FUSED TO METAL

搪瓷涂层
Vitreous Enameling

中英文对照审核稿 / Bilingual Review Draft

Primary keyword

enameling services

Content preview · Not the publish-ready WordPress version

内容预览 · 非最终 WordPress 发布版

Review Status：中英文对照审核稿 / Bilingual Review Draft
Suggested URL：`/enameling-services/`
Canonical URL Status：待确认现有站点结构 / Confirm against the existing site structure
Meta Title：Enameling Services | Vitreous Enamel for Metal Parts - Xmake
Meta Description：Xmake 为钢、铸铁及经评估的金属零件提供 enameling services，通过前处理、瓷釉施加与高温烧成形成玻璃质保护和装饰表面。提交材料、图纸、使用介质、颜色和验收要求，获取工艺评估。
Meta Description: Xmake provides enameling services for steel, cast iron, and evaluated metal parts, using preparation, enamel application, and high-temperature firing to form a vitreous protective and decorative surface. Submit the material, drawing, service media, color, and acceptance requirements for review.
Primary Keyword：enameling services
Secondary Keywords：vitreous enamel coating, porcelain enamel services, enamel coating for metal, glass fused to steel, industrial enameling, enameled steel parts, cast iron enameling
Query Variants：What is vitreous enameling? · Porcelain enamel vs enamel paint · Which metals can be enameled? · How do you prevent enamel chipping? · What causes fishscale defects in enameled steel?
Key Entities：enamel frit · ground coat · cover coat · vitreous coating · firing · enameling steel · cast iron · thermal expansion · fishscaling · edge coverage
Search Intent：交易型 / 商业调研型；买家需要判断零件能否承受搪瓷烧成、基材是否适配，并确定颜色、耐化学、边缘、冲击和外观验收要求。
Search Intent: Transactional / commercial investigation; the buyer needs to determine whether the part can tolerate enamel firing, whether the substrate is compatible, and how to specify color, chemical resistance, edges, impact, and appearance acceptance.

H1：金属零件搪瓷服务 / Enameling Services for Metal Parts

一块搪瓷表面在装配时崩边，问题往往不是从装配开始的。钢材牌号、焊缝、锐边、烧成变形、釉层体系和挂烧位置，任何一项处理不当，都可能把缺陷带到成品阶段。

Xmake 提供面向钢、铸铁及经工艺评估金属零件的 enameling services。瓷釉材料施加到清洁的金属表面后经过高温烧成，形成与基材结合的无机玻璃质层，可同时承担颜色、易清洁表面和项目定义的耐介质保护。

When an enamel surface chips during assembly, the problem often started much earlier. Steel grade, weld condition, sharp edges, firing distortion, enamel system, and support points can each carry a defect into the finished part.

Xmake provides enameling services for steel, cast iron, and process-reviewed metal parts. Enamel material is applied to a clean metal surface and fired at high temperature, producing a bonded inorganic vitreous layer that can provide color, a cleanable surface, and project-defined resistance to service media.

H2：先确认：这里的 Enameling 指什么？ / First: What Does Enameling Mean Here?

本页的 enameling 指 vitreous enamel 或 porcelain enamel，即把配制好的玻璃质瓷釉烧结到金属上。它不是常温或低温固化的 enamel paint，也不是陶瓷基体上的普通釉面。

这个区别会改变整个采购判断。玻璃质搪瓷需要金属承受烧成热循环，表面坚硬但具有玻璃材料的脆性；有机搪瓷漆的施工温度较低、韧性和修补方式不同，不能用同一规格书替代。

On this page, enameling means vitreous or porcelain enamel: a formulated glassy enamel fired onto metal. It does not mean an ambient- or low-temperature-cured enamel paint, and it is not a conventional glaze on a ceramic body.

That distinction changes the sourcing decision. Vitreous enamel requires the metal to tolerate a firing cycle; its surface is hard but retains the brittleness of glass. Organic enamel paint uses lower processing temperatures and has different flexibility and repair behavior, so the two should not share one specification.

H2：您的零件适合搪瓷吗？六项判定 / Is Your Part Suitable for Enameling? Six Checks

判定项	适合继续评估的条件	需要暂停或改方案的信号
基材	已知牌号的搪瓷用钢、合适的低碳钢或经验证的铸铁	材料牌号不明、混料、镀层或残留表面处理未说明
烧成承受能力	零件能承受瓷釉体系要求的热循环，性能与形状变化可接受	热处理状态不能改变、薄壁容易翘曲、钎焊或嵌件不耐温
几何	边缘有合理圆角，能排气排液，表面可施釉并可挂烧	刀口边、封闭夹层、深盲腔、无法支撑的大片薄板
使用载荷	主要需求是表面保护、颜色、清洁或耐介质	持续强冲击、局部弯折或装配后仍需塑性成形
验收定义	已明确接触介质、温度、可见面、允许缺陷和测试方法	只写“耐腐蚀”“食品级”或“无缺陷”，没有适用条件
后续工序	焊接、成形和大部分机加工已在施釉前完成	计划在搪瓷后焊接、攻丝、扩孔或强力压装

Check	Suitable for Further Review	Stop or Change Route When
Substrate	Known enameling steel, suitable low-carbon steel, or validated cast iron	Unknown or mixed grade, plating, or prior surface treatment is not disclosed
Firing tolerance	The part can accept the enamel-system thermal cycle and resulting property or shape change	Heat-treatment condition must remain unchanged, thin walls may warp, or brazed parts/inserts cannot take the temperature
Geometry	Edges have practical radii, liquids and gases can drain, and surfaces can be coated and supported in firing	Knife edges, sealed laps, deep blind cavities, or unsupported broad thin panels are present
Service load	Main needs are surface protection, color, cleanability, or media resistance	Repeated heavy impact, local flexing, or post-enamel plastic forming is expected
Acceptance definition	Contact media, temperature, visible surfaces, allowed defects, and test methods are known	The drawing says only “corrosion resistant,” “food grade,” or “defect free” without conditions
Downstream work	Welding, forming, and most machining are complete before enameling	Welding, tapping, reaming, or force fitting is planned after firing

判定结果 / Decision

六项中有一项未确定，不代表零件一定不能搪瓷；它意味着报价前需要材料证明、热循环样件或设计调整。若零件必须保持热处理性能、承受反复冲击或在涂层后继续大变形成形，粉末喷涂、湿式涂装或其他表面体系通常更值得先比较。

One unresolved check does not automatically disqualify the part; it means the quotation needs material evidence, a firing trial, or a design change. If the part must retain heat-treated properties, endure repeated impact, or undergo substantial forming after coating, powder coating, wet paint, or another surface system deserves comparison first.

H2：搪瓷层如何形成 / How a Vitreous Enamel Layer Is Formed

搪瓷配方以 enamel frit 为核心。熔块经研磨并配合颜料、悬浮或工艺添加组分，制成湿法釉浆或适合静电施加的干粉；实际组成要与金属、烧成制度、颜色和使用介质匹配。

施釉后的零件先干燥，再进入受控烧成。瓷釉软化、流平并在金属界面发生结合反应，冷却后成为连续玻璃质层。烧成窗口过窄、金属与釉层热膨胀不匹配、气体排出受阻或表面污染，都可能破坏最终结果。

Enamel frit is the core of the coating formulation. The frit is milled and combined with pigments, suspension agents, or process additions to make a wet slip or a dry powder suitable for electrostatic application; the composition must match the metal, firing schedule, color, and service media.

After application, the part is dried and fired under controlled conditions. The enamel softens, flows, and reacts at the metal interface, then cools into a continuous vitreous layer. A narrow or missed firing window, thermal-expansion mismatch, trapped gas, or surface contamination can compromise the result.

H2：按零件目标选择搪瓷体系 / Select the Enamel Route by Part Objective

底釉 + 面釉 / Ground Coat + Cover Coat

底釉负责与金属建立可靠结合并为面釉提供基础；面釉负责最终颜色、光泽和项目要求的表面性能。该路线适合需要分别控制基材结合与外观的钢制零件，具体采用一次或多次烧成由体系决定。

适用目标：稳定结合 · 明确颜色 · 功能与外观分层控制 需要确认：钢材牌号 · 两层兼容性 · 总厚度 · 烧成次数

The ground coat establishes dependable bonding to the metal and supports the cover coat; the cover coat controls final color, gloss, and project-required surface behavior. This route suits steel parts where substrate bonding and appearance need separate control. The system determines whether one or multiple firings are used.

Target: Stable bond · Defined color · Separate control of function and appearance Confirm: Steel grade · Interlayer compatibility · Total thickness · Number of firings

直接搪瓷体系 / Direct-on Enamel System

部分经确认的搪瓷用钢可评估直接施加面釉的工艺，减少独立底釉层或烧成步骤。能否使用 direct-on 取决于钢材化学成分、表面准备、瓷釉体系和预期性能，不能把普通钢板默认视为适用。

适用目标：经验证钢材上的简化涂层路线 需要确认：搪瓷钢规格 · 表面准备 · 试烧结果 · 覆盖与附着

Some qualified enameling steels can be evaluated for direct-on cover enamel, reducing a separate ground coat or firing step. Direct-on suitability depends on steel chemistry, preparation, enamel formulation, and expected performance; ordinary sheet steel should not be assumed compatible.

Target: Simplified coating route on validated steel Confirm: Enameling-steel specification · Surface preparation · Firing trial · Coverage and adhesion

铸铁搪瓷 / Cast Iron Enameling

铸铁件的含碳量、铸造表皮、孔隙、夹杂和局部厚度会影响放气与表面质量。适配的清理、预热或烧成路线需要通过样件建立；铸件上的孔洞和粗糙度不会被瓷釉自动修复。

适用目标：炊具部件、炉具部件、壳体或项目定义的铸铁表面 需要确认：铸铁牌号 · 铸造缺陷 · 放气 · 边缘和厚薄过渡

Carbon level, casting skin, porosity, inclusions, and local section thickness affect outgassing and surface quality on cast iron. Cleaning, preheating, and firing need a sample-based route; pores and roughness in the casting are not automatically repaired by enamel.

Target: Cookware components, heating-appliance parts, housings, or other defined cast-iron surfaces Confirm: Cast-iron grade · Casting defects · Outgassing · Edges and section transitions

特殊金属与装饰体系 / Special-Metal and Decorative Systems

铝、不锈钢、铜及其他金属存在专用瓷釉与较窄的工艺窗口，可在有明确材料和用途时评估。它们不能沿用普通钢的前处理、瓷釉或烧成参数；颜色样板也必须使用实际基材和实际烧成路线制作。

适用目标：经证实体系支持的装饰或功能表面 需要确认：精确合金 · 专用瓷釉 · 热膨胀匹配 · 颜色样件

Aluminum, stainless steel, copper, and other metals require dedicated enamels and narrower process windows and may be evaluated when the material and use are clear. They cannot inherit the pretreatment, enamel, or firing parameters used for ordinary steel; color samples must use the actual substrate and firing route.

Target: Decorative or functional surfaces supported by a validated system Confirm: Exact alloy · Dedicated enamel · Thermal-expansion match · Color sample

H2：材料不是采购清单上的普通一行 / Substrate Is Not a Generic Line Item

搪瓷用钢与低碳钢 / Enameling Steel and Low-Carbon Steel

钢板的化学成分、碳含量、脱碳状态、成形性和储氢行为会影响烧成、附着和鱼鳞爆缺陷风险。重要项目应在零件制造前锁定钢材规格；只在成品阶段补写“低碳钢”通常不够。

Sheet chemistry, carbon level, decarburization condition, formability, and hydrogen behavior influence firing, adhesion, and fishscale risk. Critical projects should lock the steel specification before part fabrication; adding “low-carbon steel” only at finished-part inspection is usually insufficient.

铸铁 / Cast Iron

铸铁能提供稳定的厚壁基体，但孔隙、石墨、夹砂、修补区和厚度变化会改变放气和热响应。应把铸件质量标准与搪瓷外观标准联动，而不是由涂层工序承担所有铸造缺陷。

Cast iron provides a stable, heavy substrate, but porosity, graphite, sand inclusions, repaired areas, and section changes alter outgassing and thermal response. Casting quality and enamel appearance standards should be linked rather than assigning every casting defect to the coating operation.

塑料与含低温嵌件组件 / Plastics and Assemblies with Low-Temperature Inserts

常规塑料不能承受玻璃质搪瓷的烧成过程，因此不属于本页 enameling services 的常规基材。塑料件需要比较 enamel paint、粉末喷涂、PVD
或其他低温表面处理；带塑料、密封件、电子元件或低熔点钎料的金属组件通常也应在搪瓷后装配。

Routine plastics cannot withstand the firing used for vitreous enamel and are outside the normal substrate scope of these enameling services. Plastic parts should be compared with enamel paint, powder coating, PVD, or another lower-temperature finish; metal assemblies containing plastics, seals, electronics, or low-melting brazes are normally assembled after enameling.

H2：图纸在施釉前要解决的八个问题 / Eight Drawing Issues to Resolve Before Coating

1. 锐边 / Sharp edges：瓷釉在刀口边难以保持稳定覆盖，且边缘更容易因冲击崩损；应评估圆角或折边设计。
2. 孔与切口 / Holes and cutouts：孔边、激光切口和冲压断面需要明确去毛刺、圆角和覆盖要求。
3. 焊缝 / Welds：焊渣、飞溅、咬边、不同填充材料和封闭气穴会影响清洁、放气和外观。
4. 封闭结构 / Closed constructions：搭接、夹层和盲腔可能滞留清洗液、气体或釉料，烧成时形成缺陷。
5. 薄壁与大平面 / Thin walls and broad panels：热循环可能导致翘曲；筋、折边、支撑方式和烧成方向应一起评审。
6. 挂点与支撑点 / Hanger and support points：工装接触区可能留下局部未覆盖或印记，应布置在非关键表面。
7. 双面施釉 / Two-sided coating：双面体系可能用于平衡覆盖或应力，但不能脱离材料、几何和工艺验证单独指定。
8. 搪瓷后加工 / Post-enamel operations：攻丝、扩孔、铆接和压装会对玻璃质层施加局部载荷；优先在施釉前完成或设计无釉功能区。

H2：颜色、光泽和表面结果 / Color, Gloss, and Surface Results

表面目标	可评估的结果	不能忽略的变量
高光纯色	平滑反光表面，颜色由面釉体系与烧成决定	基材底色、釉层厚度、烧成曲线和批次色差
哑光或半哑光	较低反射和不同触感	配方、表面微观结构、清洁方式和耐污要求
斑点或颗粒视觉	有意形成的多色或颗粒化装饰效果	图案密度、颗粒分布、主视面和限度样

表面目标	可评估的结果	不能忽略的变量
内外表面分工	内表面按介质与清洁要求选体系，外表面按颜色和环境选体系	两面材料兼容、边缘交界、总烧成路线和识别方式
工业功能面	按化学介质、温度或清洁条件开发的玻璃质表面	实际介质浓度、温度、停留时间、热循环和机械载荷

Surface Objective	Reviewable Result	Variables That Cannot Be Ignored
High-gloss solid color	Smooth reflective surface; the cover enamel and firing determine color	Substrate background, enamel thickness, firing profile, and batch color variation
Matte or semi-matte	Lower reflection and a different tactile character	Formulation, surface microstructure, cleaning method, and soil-release requirement
Speckled or particulate look	Deliberate multicolor or granular decorative appearance	Pattern density, particle distribution, primary viewing surface, and limit samples
Different inside and outside duties	Interior selected for media and cleaning; exterior selected for color and environment	Compatibility between both sides, edge transition, total firing route, and identification
Industrial functional surface	Vitreous surface developed around chemical media, temperature, or cleaning conditions	Actual concentration, temperature, dwell time, thermal cycling, and mechanical load

外观词不能替代验收标准。“白色高光”至少还需要实物色样、标准光源、可见面、允许黑点或针孔范围，以及挂点位置。功能面则应先定义介质与失效后果，再决定是否需要化学、冲击、附着或热循环测试。

Appearance words do not replace acceptance criteria. “Gloss white” still needs a physical color sample, controlled light, visible surfaces, an allowance for black specks or pinholes, and defined support points. For functional surfaces, define the media and failure consequence first, then select chemical, impact, adhesion, or thermal-cycle tests.

H2：常见失效不是同一种问题 / Common Failures Do Not Share One Cause

失效现象	常见关联因素	设计或工艺控制方向
崩边或缺口	锐边、局部冲击、压装、边缘釉层状态或热膨胀应力	增加圆角、隔离冲击、调整无釉区并做装配验证
针孔、气泡或凹坑	表面污染、基材放气、施釉状态、干燥或烧成不足	改善清洁与放气控制，确认涂布和烧成窗口
鱼鳞爆 / Fishscaling	钢中氢在冷却或储存后迁移到钢-釉界面	使用适合搪瓷的钢材，控制材料、前处理和烧成路线
露底或薄边	复杂几何、锐边、施釉不均、流动或工装遮挡	优化边缘、施釉方向、挂具和覆盖验收
颜色或光泽波动	釉层厚度、烧成温度 / 时间、基材状态和批次差异	用同基材样板、受控烧成与限度样管理
翘曲	薄壁、大平面、残余应力、支撑方式和热循环	在结构阶段增加刚度，评估工装并进行试烧测量
局部腐蚀	切边、孔隙、破损、未覆盖区或介质不匹配	明确边缘覆盖与缺陷限值，按真实介质验证完整系统

Failure Appearance	Commonly Related Factors	Design or Process Control Direction
Chipped edge	Sharp geometry, local impact, force fitting, edge enamel condition, or thermal-expansion stress	Add radii, isolate impact, revise bare zones, and validate assembly
Pinhole, bubble, or crater	Surface contamination, substrate outgassing, application condition, drying, or incomplete firing	Improve cleaning and gas control; confirm the application and firing window
Fishscaling	Hydrogen in steel migrating to the steel-enamel interface during cooling or storage	Use steel suited to enameling and control material, pretreatment, and firing route
Show-through or thin edge	Complex geometry, sharp edges, uneven application, flow, or fixture shadowing	Improve edges, application direction, fixturing, and coverage acceptance
Color or gloss shift	Enamel thickness, firing temperature/time, substrate condition, and batch variation	Manage with same-substrate samples, controlled firing, and limit samples

Failure Appearance	Commonly Related Factors	Design or Process Control Direction
Warpage	Thin walls, broad panels, residual stress, support method, and thermal cycle	Add stiffness during design, review tooling, and measure a firing trial
Local corrosion	Cut edge, pore, chip, bare zone, or media mismatch	Define edge coverage and defect limits; validate the complete system in actual media

H2：搪瓷、粉末喷涂还是湿式涂装？ / Enamel, Powder Coat, or Wet Paint?

判断维度	玻璃质搪瓷	粉末喷涂	湿式涂装
成膜材料	烧成后的无机玻璃质层	热固性或热塑性聚合物粉末	液态有机涂料
工艺温度影响	高温烧成，必须评估材料性能和变形	固化温度较低，但仍需检查基材和组件耐温	可选较低固化温度，体系范围广
表面特征	硬、玻璃质、易清洁；边缘冲击与弯折需控制	韧性和颜色 / 纹理选择通常更灵活	薄膜、修补和复杂颜色方案更灵活
适合的采购重点	介质、烧成、基材、边缘、冲击和瓷釉缺陷	前处理、膜厚、固化、边缘覆盖和耐候	底漆 / 面漆、溶剂、固化、流挂和维护
不应默认	所有金属、所有化学介质、永不崩瓷	所有锐边都有完整覆盖、不会划伤	单层就能满足全部防腐和耐化学要求

Decision Dimension	Vitreous Enamel	Powder Coat	Wet Paint
Film material	Fired inorganic glassy layer	Thermoset or thermoplastic polymer powder	Liquid organic coating
Process-temperature effect	High-temperature firing; material properties and distortion require review	Lower cure temperature, though substrate and assembly temperature limits still matter	Lower-temperature options and broad system selection are available
Surface character	Hard, vitreous, and cleanable; edge impact and bending need control	Toughness and color/texture choice are often more flexible	Thin films, repair, and complex color schemes are more flexible
Main sourcing focus	Media, firing, substrate, edges, impact, and enamel defects	Pretreatment, thickness, cure, edge coverage, and weathering	Primer/topcoat, solvent, cure, runs, and maintenance
Do not assume	Every metal, every chemical medium, or immunity to chipping	Complete sharp-edge coverage or immunity to scratching	One coat can meet every corrosion and chemical-resistance requirement

没有一种工艺在所有维度上占优。搪瓷适合把玻璃质表面特性与金属结构结合的项目；当零件需要后续变形、现场修补、较强抗冲击韧性或较低热负荷时，有机涂层通常更容易匹配。

No process leads on every criterion. Enamel suits projects that need a vitreous surface combined with a metal structure. Organic coatings are often easier to match when the part needs post-coating forming, field repair, greater impact toughness, or a lower thermal load.

H2：Xmake 搪瓷工艺路线 / Xmake Enameling Process Route

1. 用途定标 / Service Definition：记录接触介质、温度、清洁方式、冲击风险、颜色和外观等级。
2. 材料与图纸冻结 / Material and Drawing Freeze：确认金属牌号、厚度、焊接、边缘、孔、挂点和无釉区。
3. 试片或样件 / Coupon or Part Trial：使用实际基材验证前处理、瓷釉、颜色、烧成和缺陷边界。
4. 金属预处理 / Metal Preparation：去除油污、氧化皮、锈蚀和制造残留，建立适合施釉的表面。
5. 施釉与干燥 / Enamel Application and Drying：按体系采用湿法或干粉方式施加底釉、面釉或直接搪瓷层，并控制干膜状态。
6. 烧成 / Firing：按零件与瓷釉体系控制升温、产品温度、保温和冷却，使釉层熔融并与金属结合。

7. 分级检查 / Graded Inspection：区分关键功能面、主要外观面、次要表面和工装接触区，检查覆盖、颜色、针孔、崩边和变形。
8. 功能确认与包装 / Functional Confirmation and Packaging：按项目执行测试，并隔离硬接触点，避免运输过程造成边缘碰损。

H2：验收计划应从失效后果开始 / Build Acceptance Around the Failure Consequence

装饰面、炊具接触面、建筑外板和化工介质接触面不应共用一张检验表。相同的针孔在次要外表面可能属于外观分级问题，在持续接触腐蚀介质的功能面则可能不可接受。

Decorative faces, cookware-contact surfaces, architectural panels, and chemical-media contact surfaces should not share one inspection sheet. The same pinhole may be a cosmetic grading issue on a secondary exterior face and unacceptable on a functional surface continuously exposed to a corrosive medium.

验收目标	建议定义的输入	可选确认方式
外观一致性	标准样、限度样、可见面、观察条件和缺陷分级	颜色 / 光泽测量与受控目检
覆盖完整性	功能面、边缘、孔位、工装点和允许露底区	目检或与零件和釉层厚度相适配的电气孔隙 / 漏点检查
附着与抗冲击	实际基材、冲击方向、能量或变形方式	项目指定的附着、变形或冲击试验
耐化学性	介质、浓度、温度、液相 / 气相、时间和清洁循环	选定 ISO 28706 系列方法或经双方确认的介质试验
尺寸与形状	烧成前后关键尺寸、平面度、圆度和装配基准	首件测量、工装检具和试装
热循环	工作温度、升降温速度、循环次数和允许变化	与产品工况对应的热循环或热冲击方案

Acceptance Goal	Inputs to Define	Possible Confirmation Method
Appearance consistency	Master sample, limit sample, visible faces, viewing conditions, and defect grade	Color/gloss measurement and controlled visual inspection
Coverage integrity	Functional surfaces, edges, holes, fixture points, and allowed bare zones	Visual inspection or an electrical porosity / holiday test suited to the part and enamel thickness
Adhesion and impact	Actual substrate, impact direction, energy, or deformation mode	Project-specified adhesion, deformation, or impact test
Chemical resistance	Media, concentration, temperature, liquid/vapor phase, time, and cleaning cycles	Selected ISO 28706 method or mutually approved media exposure
Dimensions and shape	Critical dimensions, flatness, roundness, and assembly datums before/after firing	First-article measurement, checking fixture, and trial assembly
Thermal cycling	Working temperature, ramp rate, cycles, and allowed change	Thermal-cycle or thermal-shock plan representing the product duty

标准编号只有在适用范围与零件用途一致时才有意义。ISO 28706-2:2017 适用于玻璃质和瓷釉平面对沸腾酸、沸腾中性液体、碱性液体及其蒸气的耐化学腐蚀测定；它不是所有搪瓷零件的通用合格证，且该版本正在修订流程中。

A standard number is useful only when its scope matches the part duty. ISO 28706-2:2017 covers chemical-corrosion testing of flat vitreous and porcelain enamel surfaces against boiling acids, boiling neutral liquids, alkaline liquids, and their vapors. It is not a universal certificate for every enamel part, and this edition is in the revision process.

H2：为什么由 Xmake 协调搪瓷项目 / Why Coordinate Enameling Through Xmake

搪瓷不是把已完成零件交给涂层工序就能独立解决的问题。Xmake 可把材料采购、钣金或铸造、焊接、边缘处理、施釉、烧成和检验放进同一制造计划，先处理会在炉内暴露的风险。

- 图纸端处理烧成风险：在制造前确认圆角、排液、挂点、无釉区和变形控制。
- 同基材样件：颜色和性能样件使用实际金属与实际工艺路线，避免只看独立瓷釉色板。
- 制造与表面责任连贯：铸造孔隙、焊接残留、前处理、施釉和烧成问题可沿工艺路线追溯。
- 质量管理：Xmake 已通过 ISO 9001 和 ISO 13485 认证；这些是公司级质量管理资质，不代表某一搪瓷零件自动满足食品接触、卫生或其他受监管要求。

Enameling is not a stand-alone fix applied after the part is complete. Xmake can place material purchasing, sheet-metal fabrication or casting, welding, edge preparation, enamel application, firing, and inspection in one manufacturing plan and address risks that will surface in the furnace.

- Address firing risk in the drawing: Confirm radii, drainage, support points, bare zones, and distortion control before fabrication.
- Sample on the actual substrate: Produce color and performance samples with the real metal and route rather than relying on an isolated enamel chip.
- Maintain process accountability: Trace casting pores, weld residue, preparation, application, and firing issues through the route.
- Quality management: Xmake is ISO 9001 and ISO 13485 certified. These are company-level quality management credentials and do not mean that an individual enameled part automatically meets food-contact, sanitary, or other regulated requirements.

H2：常见问题 / Frequently Asked Questions

Porcelain enamel、vitreous enamel 和 enamel paint 是一回事吗？ / Are Porcelain Enamel, Vitreous Enamel, and Enamel Paint the Same?

Porcelain enamel 与 vitreous enamel 通常指烧成到金属上的玻璃质涂层。Enamel paint 是有机涂料名称，不经过相同的玻璃熔融和金属结合过程，材料、温度与性能边界不同。

Porcelain enamel and vitreous enamel normally refer to a glassy coating fired onto metal. Enamel paint is an organic coating name and does not undergo the same glass fusion and metal-bonding process; its material, temperature, and performance boundaries differ.

搪瓷能用于不锈钢或铝吗？ / Can Stainless Steel or Aluminum Be Enameled?

可以评估，但必须使用与合金和热膨胀特性匹配的专用体系。普通钢的瓷釉、前处理和烧成路线不能直接复制到不锈钢或铝上。

They can be evaluated, but a dedicated system matched to the alloy and thermal-expansion behavior is required. An ordinary-steel enamel, preparation route, and firing schedule cannot be copied directly to stainless steel or aluminum.

搪瓷为什么会崩边？ / Why Does Enamel Chip at an Edge?

常见关联因素包括锐边、局部厚釉或薄釉、热膨胀应力、工装接触和装配冲击。改进通常从零件圆角、无釉区、工装和装配方式入手，而不是只增加釉层厚度。

Commonly related factors include sharp edges, locally thick or thin enamel, thermal-expansion stress, fixture contact, and assembly impact. Improvement usually starts with part radii, bare zones, tooling, and assembly method rather than simply adding enamel thickness.

鱼鳞爆缺陷会立即出现吗？ / Does Fishscaling Always Appear Immediately?

不一定。鱼鳞爆与钢中的氢迁移到钢-釉界面有关，可能在冷却后或储存期间出现。材料选择、储氢行为、前处理和烧成控制都需要纳入预防方案。

Not always. Fishscaling is associated with hydrogen migrating to the steel-enamel interface and may appear after cooling or during storage. Material selection, hydrogen-trapping behavior, pretreatment, and firing control all belong in the prevention plan.

询价时需要提供哪些资料？ / What Should Be Included in an Enameling RFQ?

请提供图纸、金属牌号与材料证明、数量、焊接和前处理状态、目标颜色 / 光泽、接触介质、工作温度、关键表面、允许无釉区、冲击或热循环要求，以及适用标准。已有不良样件或限度样时，请一并提供。

Provide the drawing, metal grade and material evidence, quantity, weld and pretreatment condition, target color/gloss, contact media, operating temperature, critical surfaces, allowed bare zones, impact or thermal-cycle requirements, and applicable standards. Include failed parts or limit samples when available.

CTA：获取搪瓷涂层项目评估 / Request an Enameling Project Review

上传零件图纸并标注材料、厚度、焊缝、边缘、挂点限制、功能面和外观面。Xmake 将评估基材、几何、瓷釉体系、烧成风险、样件计划和验收路径，再确认报价所需信息。

Upload the part drawing and identify material, thickness, welds, edges, support-point restrictions, functional surfaces, and cosmetic faces. Xmake will review the substrate, geometry, enamel system, firing risks, sample plan, and acceptance route before confirming quotation inputs.

建议内部链接 / Suggested Internal Links

- Xmake Surface Finishing Services / Xmake 表面处理服务
- Sheet Metal Fabrication Services / 钣金加工服务
- Metal Casting Services / 金属铸造服务
- Powder Coating Services / 粉末喷涂服务
- Electrophoretic Coating Services / 电泳涂装服务
- Request a Quote / 获取报价

人工核验说明 / Human Verification Notes

- Needs verification：实际金属牌号是否适合目标瓷釉、前处理和烧成路线，特别是 direct-on、铸铁、铝、不锈钢或其他特殊体系。
- Needs verification：零件在烧成后的材料性能、平面度、圆度、关键尺寸与装配要求。
- Needs verification：底釉 / 面釉或直接搪瓷体系、颜色、光泽、总厚度、烧成次数和可获得性。
- Needs verification：关键面、边缘、孔、挂点、无釉区、允许针孔 / 黑点 / 露底 / 色差与限度样。
- Needs verification：接触介质、浓度、温度、时间、清洁循环、冲击、热循环及相应测试方法和判定值。
- Needs verification：食品接触、饮用水、卫生、低析出、法规、法定认证或行业专用要求；不得从公司 ISO 资质推导。
- Needs verification: Suitability of the actual metal grade for the target enamel, pretreatment, and firing route, especially for direct-on, cast iron, aluminum, stainless steel, or another special system.
- Needs verification: Post-firing material properties, flatness, roundness, critical dimensions, and assembly requirements.
- Needs verification: Ground/cover or direct-on system, color, gloss, total thickness, number of firings, and availability.
- Needs verification: Critical faces, edges, holes, support points, bare zones, allowed pinholes/black specks/show-through/color variation, and limit samples.

- Needs verification: Contact medium, concentration, temperature, time, cleaning cycles, impact, thermal cycles, and corresponding test methods and acceptance values.
- Needs verification: Food contact, potable water, hygiene, migration, regulatory, statutory certification, or industry-specific requirements; these cannot be inferred from company ISO credentials.

技术依据 / Technical Reference Basis

- `.agents/skills/xmake-plastichubs-website-ops/references/surface-finishing-knowledge-base.md`：将搪瓷限定为以金属、钢和铸铁为主的工艺，并要求谨慎处理塑料兼容性。
- `.agents/skills/xmake-plastichubs-website-ops/references/manufacturing-claims-control.md`：限定能力、认证、精确指标和受监管用途表述。
- Internal — Not for Publication：International Enamellers Institute：支持玻璃质搪瓷是施加到金属并经高温玻化烧成的无机涂层，主要基材包括钢和铸铁。
- Internal — Not for Publication：ASTM A424/A424M：支持瓷釉用钢板需要专门的化学成分、加工与力学性能要求；项目引用前应确认当前有效版本及适用范围。
- Internal — Not for Publication：ISO 28706-2:2017：支持特定沸腾酸、中性液体、碱性液体及蒸气条件下的耐化学腐蚀测试；该版本正在修订流程中。
- Internal — Not for Publication：Surface and Coatings Technology, 2020：支持钢板搪瓷鱼鳞爆与氢在钢-釉界面的积聚有关。
- Internal — Not for Publication：U.S. EPA Porcelain Enameling Development Document：支持金属准备、瓷釉制备与施加、干燥和烧成的基本工艺链。
- 本稿未承诺固定烧成温度、瓷釉厚度、颜色范围、耐介质等级、冲击值、交期或保证性寿命。
- `.agents/skills/xmake-plastichubs-website-ops/references/surface-finishing-knowledge-base.md`: Limits enameling mainly to metal, steel, and cast iron and requires caution around plastic compatibility.
- `.agents/skills/xmake-plastichubs-website-ops/references/manufacturing-claims-control.md`: Controls capability, certification, exact metrics, and regulated-use language.
- Internal — Not for Publication: International Enamellers Institute: Supports vitreous enamel as an inorganic coating applied to metal and vitrified by high-temperature firing, primarily on steel and cast iron.
- Internal — Not for Publication: ASTM A424/A424M: Supports dedicated chemistry, processing, and mechanical-property requirements for sheet steel intended for porcelain enameling; confirm the active version and scope before project citation.
- Internal — Not for Publication: ISO 28706-2:2017: Supports chemical-corrosion testing under specified boiling-acid, neutral-liquid, alkaline-liquid, and vapor conditions; this edition is in revision.
- Internal — Not for Publication: Surface and Coatings Technology, 2020: Supports the relationship between fishscaling and hydrogen accumulation at the steel-enamel interface.
- Internal — Not for Publication: U.S. EPA Porcelain Enameling Development Document: Supports the basic process chain of metal preparation, enamel preparation/application, drying, and firing.
- This draft does not promise a fixed firing temperature, enamel thickness, color range, media-resistance class, impact value, lead time, or guaranteed service life.